

13. Two solids 'X' and 'Y' can be separated by using water as a solvent. Identify X and Y.

- (a) NaCl and P
- (b) S and C
- (c) NaCl and C
- (d) NaCl and Sugar

14. **Assertion (A):** On opening a Pepsi bottle, CO_2 comes out with fizzing sound.

Reason (R): Solubility of a gas is inversely proportional to pressure.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

15. Which among the following applications is not based on the principle of fractional distillation?

- (a) Production of petrol from crude oil
- (b) Making of coffee decoction
- (c) Purification of ethyl alcohol
- (d) Separation of acetone and water

Space for rough work

Assessment Test III

Time: 30 min

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Wet clothes dry faster on a windy day.

Reason (R): Rate of evaporation is proportional to humidity.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

2. **Assertion (A):** Compounds are invariably homogeneous.

Reason (R): Compounds are constituted of only one kind of molecules distributed uniformly.

- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true and R is false.
- (d) A is false and R is true.

3. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Solvent extraction	(A) Difference in densities
(ii) Gravity method	(B) Size of the particles of insoluble solid
(iii) Sublimation	(C) High density of insoluble solid
(iv) Fractional crystallization	(D) Solubility of one component in a solvent
	(E) Ability of one component to sublime
	(F) Difference in solubility of the components in the same solvent at a certain temperature range

- (a) (i) → (D); (ii) → (A); (iii) → (E); (iv) → (F)
- (b) (i) → (D); (ii) → (F); (iii) → (E); (iv) → (A)
- (c) (i) → (B); (ii) → (C); (iii) → (E); (iv) → (F)
- (d) (i) → (D); (ii) → (A); (iii) → (E); (iv) → (C)

4. Arrange the following steps in a proper sequence for the separation of a mixture of kerosene and water.

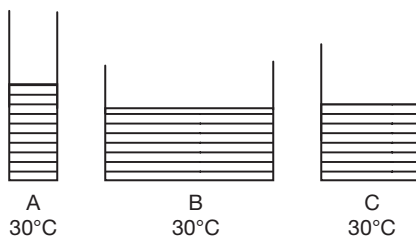
- (A) The mixture is kept standstill.
- (B) The liquid mixture is taken in a separating funnel.

Space for rough work

- (C) The heavier component is first collected by opening the stop lock.
(D) Clear layers of the liquids are formed and the heavier liquid that is water forms the bottom layer.
- (a) BADC (b) BCAD
(c) BCDA (d) ADCB
5. Identify the wrong statement among the following.
- (a) In solids, molecules are closely packed.
(b) In solids, molecules have very strong intermolecular forces of attraction.
(c) In solids, molecules have only vibratory motion about their mean positions.
(d) In solids, molecules possess very high kinetic energy.
6. The various steps involved in the process of condensation are given below. Arrange them in a proper sequence.
- (A) Further extraction of heat results in the decrease in potential energy of molecules, whereas kinetic energy remains constant.
(B) The gas molecules, on cooling, lose heat energy which decreases the kinetic energy as well as potential energy of the molecules.
(C) Increase in the forces of attraction results in the conversion of state.
(D) Decrease in potential energy leads to increase in intermolecular forces of attraction.
- (a) ABCD (b) BADC (c) BACD (d) ABDC
7. Identify the correct statement among the following.
- (a) Boiling point of liquid decreases with increase in external pressure.
(b) Boiling point of liquid decreases by adding impurities.
(c) Addition of impurities to the solids decreases the melting point of solids.
(d) Addition of impurities to the solids increases the melting point of solids.
8. Conduction of heat is not possible through gases. Why?
- (a) Gas molecules are very loosely packed.
(b) Gas molecules possess vibratory, rotatory and translatory motions.
(c) Gas molecules possess very high kinetic energy.
(d) Gas molecules are highly diffusible.
9. In which among the following cases is evaporation not involved?
- (a) Skin becomes rough in winter.
(b) An earthen pot keeps water cool.
(c) In early winter morning, exhaled air appears foggy.
(d) Cool sensation is perceived when perfume is sprayed on the palms.

10. Which of the following mixtures can be separated by the process of magnetic separation?
- (a) Chalk powder + sand (b) Iron + sand
(c) Sulphur + calcium (d) Common salt + sand
11. In which among the following cases is adhesive force predominant over cohesive force?
- (a) Water in capillary tube.
(b) Mercury in capillary tube.
(c) Water in a test tube in which the inner wall is coated with wax.
(d) Water on leaves of aquatic plants.
12. Which of the following gaseous mixtures can be separated by using KOH as a solvent?
- (a) CO_2 and SO_2 (b) N_2 and NH_3
(c) CO_2 and NH_3 (d) H_2 and N_2
13. Four gases A, B, C, and D have critical temperatures 32°C , -118°C , 31.1°C , and -140°C , respectively. Arrange them in the increasing order of ease of liquefaction of the gases.
- (a) $\text{D} < \text{B} < \text{C} < \text{A}$ (b) $\text{A} > \text{B} > \text{D} > \text{C}$
(c) $\text{C} < \text{D} < \text{B} < \text{A}$ (d) $\text{A} < \text{C} < \text{B} < \text{D}$
14. An increase in pressure increases the melting point. This statement is true for _____.
- (a) ice and brass (b) silver and gold
(c) brass and gold (d) ice and silver
15. Which among the following disappears with time without heating?
- (a) Ammonium chloride (b) Naphthalene
(c) Sodium chloride (d) Sulphur

5. Which among the following possesses maximum intermolecular force of attraction?
- (a) Bromine (b) Air
(c) Oxygen (d) Copper
6. The steps involved in the formation of dew are given below. Arrange them in a proper sequence.
- (A) On further lowered in temperature, water vapour condenses.
(B) During day time water from water bodies gets evaporated.
(C) Temperature falls during night time, and hence the air becomes saturated with water vapour.
- (a) CBA (b) CAB (c) BCA (d) ACB
7. Pressure cooker reduces the cooking time. Choose the correct reason from given options.
- (a) Pressure inside the cooker is more, so boiling point of water decreases.
(b) Pressure inside the cooker is less, so boiling point of water decreases.
(c) Pressure inside the cooker is more, so boiling point of water increases.
(d) Generally, cookers are made up of aluminium metal, which is a good conductor of heat.
8. Maximum fluidity of gases is due to _____.
- (a) high intermolecular space.
(b) high kinetic energy of molecules.
(c) the presence of three types of molecular movements.
(d) the absence of free surfaces.
9. Three liquids are taken in three different types of containers. What is the correct order of the rate of evaporation?



- (a) $A < B < C$ (b) $A > C > B$
(c) $B > C > A$ (d) $C < A < B$

Space for rough work

10. Mixture of NaCl and NH_4Cl can be separated by ____.
- (a) gravity method
 - (b) solvent extraction
 - (c) sublimation
 - (d) fractional crystallization
11. Cotton clothes are preferred in summer. Which among the following is the reason for this statement?
- (a) They are attractive and cheaper.
 - (b) Cotton clothes absorb sweat more due to the greater adhesive forces between cotton and water.
 - (c) Cotton is an artificial fibre.
 - (d) Cotton fibres do not absorb sweat.
12. Which among the following gaseous mixtures can be separated by fractional evaporation?
- (a) $\text{H}_2 + \text{SO}_2$
 - (b) $\text{CO} + \text{CO}_2$
 - (c) $\text{NH}_3 + \text{SO}_2$
 - (d) $\text{O}_2 + \text{N}_2$
13. Identify the false statement regarding critical temperature.
- (a) It is directly proportional to boiling point.
 - (b) Every gas is associated with a specific critical temperature.
 - (c) It is directly proportional to intermolecular forces.
 - (d) A gas cannot be liquefied below its critical temperature.
14. Addition of impurities to a solid decreases the melting point. In which among the following this phenomenon is not exploited?
- (a) Fish and meat can be preserved for a longer time in ice if common salt is added to it.
 - (b) Brine solution is used to remove the ice piled on roads in cold regions.
 - (c) Melting point of rose metal is less than that of its components.
 - (d) Regelation
15. Solid carbon dioxide undergoes sublimation due to _____.
- (a) high intermolecular forces of attraction
 - (b) high kinetic energy of all molecules
 - (c) high kinetic energy of surface molecules
 - (d) high potential energy of molecules

Answer Keys

Assessment Test I

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|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (a) | 2. (c) | 3. (a) | 4. (c) | 5. (c) | 6. (b) | 7. (a) | 8. (d) | 9. (c) | 10. (c) |
| 11. (c) | 12. (c) | 13. (b) | 14. (d) | 15. (b) | | | | | |

Assessment Test II

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|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (a) | 2. (c) | 3. (a) | 4. (c) | 5. (a) | 6. (d) | 7. (a) | 8. (a) | 9. (d) | 10. (c) |
| 11. (d) | 12. (a) | 13. (c) | 14. (c) | 15. (b) | | | | | |

Assessment Test III

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|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (c) | 2. (a) | 3. (a) | 4. (a) | 5. (d) | 6. (b) | 7. (c) | 8. (a) | 9. (c) | 10. (b) |
| 11. (a) | 12. (c) | 13. (a) | 14. (b) | 15. (b) | | | | | |

Assessment Test IV

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|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (a) | 2. (a) | 3. (c) | 4. (b) | 5. (d) | 6. (c) | 7. (c) | 8. (b) | 9. (c) | 10. (c) |
| 11. (b) | 12. (d) | 13. (d) | 14. (d) | 15. (c) | | | | | |

Atomic Structure

2



Reference: Coursebook - IIT Foundation Chemistry Class 9; Chapter - Atomic Structure; Page number - 2.1–2.14

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

- Which of the following is the correct electronic configuration of an element with atomic number 29?
(a) 2, 8, 18, 1 (b) 2, 8, 17, 2 (c) 2, 9, 16, 2 (d) 2, 18, 9
- Identify the element from the given configuration.
 $a, 4a, (6a + 1)$, and $(a - 1)$
(a) Scandium (b) Chromium
(c) Iron (d) Manganese
- Which of the following postulates of Dalton's atomic theory was proved wrong with the discovery of radioactivity?
(a) Matter is constituted of the tiniest particles called atoms.
(b) Atoms are indivisible.
(c) Atoms of same elements are identical in all aspects.
(d) Atoms of different elements are different in all aspects.
- The ratio of atomic numbers of two elements 'X' and 'Y' is 1:7. The radius of the third orbit of single electron species of X is $1.587 \text{ \AA}$. Identify the penultimate shell of Y.
(a) N-shell (b) M-shell
(c) L-shell (d) O-shell
- A dipositive ion of element 'A' is isoelectronic with inert gas 'X'. X possesses three completely filled shells. The number of electrons present in the antepenultimate shell of atom of element A is _____.
(a) 18 (b) 8
(c) 32 (d) 2
- According to Rutherford's atomic model, _____.
(a) protons are the lightest positively charged particles
(b) electrons rotate in the opposite spin to the rotation of protons
(c) electrons revolve around the nucleus in certain fixed paths
(d) electrons revolve around the nucleus with high velocity

7. In α -ray scattering experiment, _____.
- α -rays were deviated from their path due to attraction by the electrons
 - gold foil is used since the number of electrons are more in gold atoms
 - gold foil is used to prevent the penetration of α -particles
 - α -particles are used because they are positively charged and have mass
8. Two isotopes 'X' and 'Y' are present in a natural sample in 3:2 ratio. If the average atomic mass is $(A + 0.8)$ and the atomic mass of the lighter isotope, that is, X is A, then calculate the atomic mass of the heavier isotope.
- $A + 1$
 - $A + 2$
 - $A + 0.5$
 - $A + 1.5$
9. Calculate the radius of the third orbit of a single electron species, if the energy of this orbit is -13.6 eV.
- $1.58 A^\circ$
 - $0.529 A^\circ$
 - $1058 A^\circ$
 - $2096 A^\circ$
10. If the energy of the second orbit of a single electron species is $-x$ eV, then calculate the atomic number of that atom in terms of x .
- $2x\sqrt{\frac{1}{13.6}}$
 - $2\sqrt{\frac{x}{13.6}}$
 - $\sqrt{\frac{2x}{13.6}}$
 - $\frac{2x}{13.6}$
11. Arrange elements 'A', 'B', 'C' and 'D' in the ascending order of their atomic numbers based on the description given below.
- A has 3 electrons in O shell.
 - B is an element with 5 electrons in the fifth shell.
 - C is a halogen with 5 shells.
 - D attains a pseudo inert gas configuration (18 electrons in the outermost shell) after losing 2 electrons from N shell.
- DBCA
 - DCBA
 - DABC
 - DACB
12. Arrange the elements 'A', 'B', 'C' and 'D' in the ascending order of the number of valence electrons.
- A is the noble gas with the least number of shells.
 - The number of electrons in B is 4 times that in A.
 - C has 7 electrons more than that in B.
 - D has 4 electrons less than that in C.
- BADA
 - DACB
 - CDBA
 - DCBA
13. **Assertion (A):** An element with atomic number 36 is unreactive.
- Reason (R):** 8 electrons in the outermost shell make the atom highly energetic.
- Both A and R are true and R is the correct explanation of A.
 - Both A and R are true, but R is not the correct explanation of A.
 - A is true and R is false.
 - A is false and R is true.